

EMMANUEL BUHARI

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Education

University of Calgary

Expected May 2028

Bachelor of Science, Computer Science

Calgary, AB

- GPA: 3.64/4.0
- Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Operating Systems, Functional Programming

Experience & Leadership

UI/UX Designer

January 2021 – October 2022

AARODS Homes & Constructions

Lagos, Nigeria

- Developed functional prototypes using **Flutter** to deliver MVPs for client review, enabling rapid iteration based on user feedback
- Designed high-fidelity wireframes and interactive mockups in **Adobe XD**, translating stakeholder requirements into polished user interfaces

Workshop Executive

September 2025 – Present

AI Club, University of Calgary

Calgary, AB

- Delivered technical workshops on **AI Agents** and **Retrieval-Augmented Generation (RAG)** to 100+ students, covering **LLM orchestration**, **tool use**, **retrieval pipeline design**, and **prompt engineering**
- Designed curriculum translating research-level concepts (ReAct agents, semantic retrieval, embedding pipelines) into hands-on implementations, with live coding demos in **Python**

Projects

Deep Notes | [Devpost](#) | [Source Code](#) | [Website](#)

TypeScript | React | RAG | Gemini | Ollama | Obsidian API

- Architected a full **RAG pipeline** in **TypeScript**: context-aware note chunking enriched with Obsidian metadata (outbound links, tags, frontmatter), **embedding generation**, and **cosine similarity search** at query-time to retrieve relevant chunks for **LLM** prompting
- Reduced vector search memory by **4x** and similarity compute by **4x** by replacing a third-party library with a custom **int8-quantized** local store using random orthogonal projection over 768-dimensional embeddings
- Improved retrieval quality by **40%** by combining **BM25** term-frequency ranking with **dense vector search** via **Reciprocal Rank Fusion** with cross-note link boosting, with automatic fallback to float-precision scoring when quantized vectors are unavailable
- Cut redundant **embedding API** calls to zero by building **incremental re-indexing** with per-chunk text hashing and a debounced file-modify listener, keeping the index current in real time without full re-indexing on every file save
- Built an **Obsidian** plugin in **TypeScript** integrating **Gemini** and local **Ollama** via a unified provider interface, enabling full offline inference without any code changes to the calling layer; submitted to the official **Obsidian** plugin registry, currently under review

Inventory Assist | [Source Code](#)

React | Node.js | Express | MySQL | JWT

- Engineered a full-stack inventory management system with dual-role access (Admin and Stock Clerk) using **React**, **Node.js/Express**, and **MySQL**, contributing **25 commits** as a primary full-stack developer
- Built complete **REST API** backend with **11 route modules** covering authentication (**JWT**), CRUD operations, purchase orders, reorder suggestions, reporting, and **Stripe** billing integration
- Developed Admin portal with bulk **CSV/XLSX** product import, 3-step purchase order workflow, inventory reports with bar charts, and role-based user management

Technical Skills

- **Languages:** Python, Java, C, JavaScript, TypeScript, SQL, HTML/CSS
- **Frameworks:** React, Node.js, Express, JavaFX, FastAPI, Flutter, spaCy, Tailwind CSS
- **Developer Tools:** Git, GitHub, VS Code, IntelliJ, Linux, MySQL